

- Numpy package helps to power N-dimensional array computing.
- It's often used in IoT to read sensor bulk data from the system's database inherent functions.
- Reference: <https://pypi.org/project/numpy/>
- **Pandas:** Install with 'pip install pandas'.
 - Pandas is a Python module that provides quick, versatile, and expressive data structures that are intended to make working with 'relational' or 'labelled' data simple and natural.
 - It aspires to be the basic high-level building block for doing realistic, real-world data analysis in Python.
 - Reference: <https://pypi.org/project/pandas/>
- **Matplotlib:** Install using 'pip install matplotlib'.
 - Matplotlib is a comprehensive library for creating static, animated, and interactive visualisations in Python.
 - Reference: <https://pypi.org/project/matplotlib/>
- **JSON**
 - JSON is a data storage and exchange syntax.
 - It has been written in JavaScript object notation.
 - Python includes a library named json that may be used to work with JSON data.
 - Reference: <https://docs.python.org/3/library/json.html>
- **IoT machine learning and deep learning (AI)**
 - **Tensorflow:** Install with 'pip install tensorflow'.
 - TensorFlow is a high-performance numerical computing open source software package.
 - Its adaptable architecture enables simple computing deployment over a wide range of platforms (CPUs, GPUs, TPUs), from PCs and clusters of servers to mobile and edge devices for handling non-linear data sets.
 - Reference: <https://pypi.org/project/tensorflow/>
 - **Opencv:** Install via 'pip install opencv-python'.
 - OpenCV is a large open source library for computer vision, machine learning, and image processing, and it currently plays an essential part in real-time operations.
 - Reference: <https://pypi.org/project/opencv-python/>
 - **Pycaret:** Install using 'pip install pycaret'.
 - PyCaret is an open source, low-code machine learning library in Python.
 - When compared to other open source machine learning libraries, PyCaret is a low-code library that may be used to replace hundreds of lines of code with only a few lines. Experiments become significantly faster and more efficient as a result, making it apt to use for IoT apps.
 - Reference: <https://pypi.org/project/pycaret/>
 - **LightGBM:** Install using 'pip install lightgbm'.
 - LightGBM is an abbreviation for Light Gradient Boosting Machine. It provides plenty of memory-efficient yet quick computational power and is capable of handling massive volumes of data.
 - Reference: <https://pypi.org/project/lightgbm/>
- **IoT cloud integration**
 - **Azure-CLI:** Install using 'pip install azure-cli'.
 - Azure-CLI provides Python tools for Azure.
 - Reference: <https://pypi.org/project/azure-cli/>
 - **AWS:**
 - Client devices can interface with AWS IoT and AWS IoT Greengrass core devices using the Python programming language by using the AWS IoT Device SDK for Python.
 - Reference: <https://docs.aws.amazon.com/greengrass/v1/developerguide/IoT-SDK.html>

Challenges of using Python for IoT

Python has many advantages, but no programming language is flawless.

- Python is slower than other programming languages because it is dynamically typed and run line by line.
 - Python's key limitations are its runtime speed, memory management, and concurrency and parallelism support. When compared to other languages, the runtime speed is poor. Due to the amount of memory Python needs, projects with many objects active in RAM may encounter challenges when using Python.
 - While Python is a good server-side language, it is rarely used on the client-side as working with mobile applications and apps is slow and inconvenient.
- IoT is evolving, and the choice of tools to meet its demands is crucial. Several programming languages have proven effective in IoT development. Python has shown itself to be a valuable tool for prototyping, developing, and operating various IoT devices and systems. Its rapid development speed, minimal learning curve, and extensive library set make it indispensable for IoT. 

References

- <https://www.datamation.com/wp-content/uploads/2021/04/IOT-4.png>
- <https://www.deepseadev.com/en/blog/iot-with-python-reasons-to-use-it/>
- <https://edu.varistor.in/python-in-iot/>
- <https://www.oodlestechnologies.com/blogs/role-of-python-in-iot-development/>
- <https://svitla.com/blog/internet-of-things-with-python>

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The author works in a leading bank as AVP. She is a perennial learner and loves the quote: "The more I learn, the more I realise how much I don't know." -Albert Einstein